

VGGNet-16 Based Blood Cell Classification on the BCCD Dataset

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Abstract—This paper presents a PyTorch implementation of VGGNet-16 with transfer learning for the classification of white blood cells using the Blood Cell Images (BCCD) dataset. The dataset comprises 12,500 microscopic images spanning four cell types: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. A stratified split of 70% training, 15% validation, and 15% testing is applied. The model achieves 91.63% test accuracy, 0.9160 macro F1-score, and 0.9857 macro AUC-ROC. Transfer learning from ImageNet-pretrained weights, combined with data augmentation, cosine learning rate annealing, and label smoothing, enables strong classification performance with a frozen convolutional backbone.

Index Terms—VGGNet-16, blood cell classification, transfer learning, deep learning, convolutional neural networks, BCCD dataset, medical image analysis.

I. INTRODUCTION

Convolutional neural networks (CNNs) remain among the most effective model families for image recognition. Modern pretrained architectures can be adapted to domain-specific problems with limited data through transfer learning, substantially reducing training time and data requirements. VGGNet-16 is a widely adopted architecture that provides a deep but interpretable convolutional backbone well-suited to fine-grained visual tasks.

Automatic blood cell classification is a clinically relevant problem. Manual examination of microscopic blood smears is time-consuming and subject to inter-observer variability. A robust automated classifier can assist pathologists in screening for conditions such as leukemia, anemia, and bacterial infections.

In this assignment, VGGNet-16 is implemented in PyTorch with ImageNet pretrained weights and evaluated on the BCCD dataset using a full metric suite including accuracy, F1-score, AUC-ROC, and confusion matrix, following strict train/validation/test separation.

The main contributions of this work are as follows:

- A complete PyTorch implementation of VGGNet-16 with a custom 4-class classifier head trained using transfer learning.
- A reproducible data pipeline with augmentation, cosine LR scheduling, label smoothing, and early stopping.
- A strict evaluation pipeline reporting accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrix.

II. RELATED WORK

VGGNet-16, introduced by Simonyan and Zisserman [1], demonstrated that depth using small 3×3 convolution filters is a critical component of strong visual recognition performance. Subsequent architectures such as ResNet [2] and EfficientNet extended this idea with skip connections and compound scaling.

Transfer learning from large-scale datasets has been

shown to be highly effective for medical imaging tasks where labeled data is scarce [3]. Surveys by Litjens et al. [4] and Shen et al. [5] summarize how CNNs have become central to classification, detection, and segmentation workflows in medical imaging.

For hematology specifically, Matek et al. [6] demonstrated CNN-based recognition of blast cells approaching expert-level performance. Public blood-cell datasets such as the BCCD dataset [7] and the peripheral blood cell dataset described by Acevedo et al. [8] have helped standardize benchmarking. Data augmentation [9] and ROC-based evaluation [10] are standard tools in this domain.

III. METHODOLOGY

A. Dataset

The BCCD dataset [7] contains 12,500 microscopic blood-smear images distributed across four white blood cell classes: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. Images are RGB JPEG files captured at $400\times$ magnification with Giemsa staining. The dataset is balanced across classes, making it suitable for standard multi-class classification.

Figure 1 shows representative samples from the dataset after augmentation.

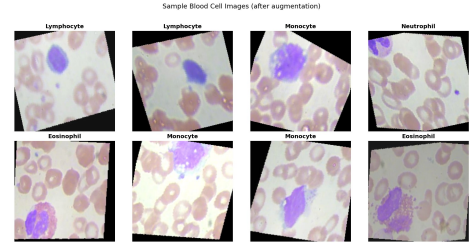


Fig. 1. Sample BCCD images after augmentation, all four cell types shown.

B. Stratified Split

A stratified random split with seed 42 is applied, yielding 6,971 training images (70%), 1,493 validation images (15%), and 1,493 test images (15%). Table I summarizes the split.

TABLE I
Dataset Split Summary

Split	Total	Per Class	Approx.
Train	6,971	~1,743	70%
Validation	1,493	~373	15%
Test	1,493	~373	15%
Total	9,957	~2,489	100%

C. VGGNet-16 Architecture

VGGNet-16 consists of 13 convolutional layers in five blocks, followed by fully connected layers, originally outputting 1,000 ImageNet classes. For this task, the final classifier is replaced with a 4-class head:

$Linear(25088 \rightarrow 4096) \rightarrow ReLU \rightarrow Dropout(0.5) \rightarrow$
 $Linear(4096 \rightarrow 1024) \rightarrow ReLU \rightarrow Dropout(0.4) \rightarrow$

The convolutional feature extractor is initialized with ImageNet pretrained weights and frozen during training. The total number of trainable parameters is approximately 26 million (classifier head only), compared to 138 million total parameters.

D. Training Configuration

The model is implemented in PyTorch and trained on a single NVIDIA T4 GPU via Google Colaboratory. Key hyperparameters are: batch size 32, Adam optimizer with learning rate 1×10^{-4} and weight decay 1×10^{-6} , cosine annealing LR scheduler ($T_{\max}=15$, $\eta_{\min}=1 \times 10^{-6}$), cross-entropy loss with label smoothing ($\epsilon=0.1$), and early stopping with patience 5. Training augmentations include random horizontal flip, random vertical flip, random rotation ($\pm 15^\circ$), and color jitter.

E. Evaluation Metrics

The model is evaluated using: Accuracy, Macro Precision, Macro Recall, Macro F1-score, Macro one-vs-rest AUC-ROC, and Confusion Matrix. Macro averages are reported to give equal weight to all four classes.

IV. EXPERIMENTS

The full pipeline consists of four stages: data preprocessing, augmentation, VGGNet-16 training with transfer learning, and final test evaluation. Training and validation curves are shown in Figure 2. The confusion matrix and ROC curves are shown in Figures 3 and 4 respectively.

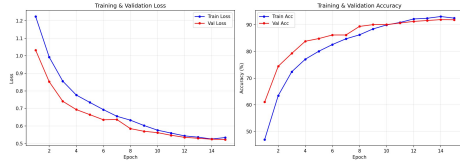


Fig. 2. Training and validation loss and accuracy curves over 15 epochs.

Table II reports the aggregate test performance. Table III reports the per-class breakdown.

TABLE II
Overall Test Performance

Metric	Value
Accuracy	91.63%
Macro Precision	87.00%
Macro Recall	91.63%
Macro F1-score	91.60%
Macro AUC-ROC (OvR)	98.57%

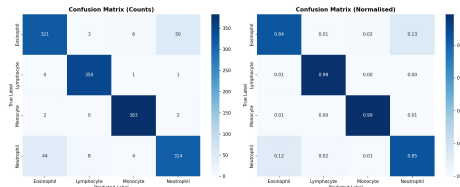


Fig. 3. Confusion matrix on the held-out test set.

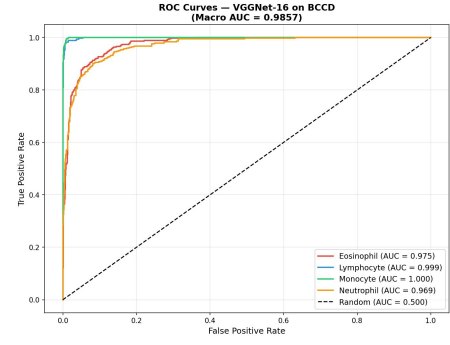


Fig. 4. One-vs-rest ROC curves for all four blood-cell classes.

TABLE III
Per-Class Test Performance

Class	Prec.	Recall	F1	AUC
Eosinophil	0.87	0.84	0.85	0.975
Lymphocyte	0.97	0.98	0.97	0.999
Monocyte	0.97	0.99	0.98	1.000
Neutrophil	0.85	0.85	0.85	0.969

The confusion matrix shows that Lymphocyte and Monocyte are classified with near-perfect accuracy (98% and 99% recall). The primary source of error is confusion between Eosinophil and Neutrophil, with 50 Eosinophils misclassified as Neutrophils (13%) and 44 Neutrophils as Eosinophils (12%), consistent with their morphological similarity.

V. DISCUSSION

The results show that VGGNet-16 with transfer learning is highly effective for this blood cell classification task. The macro AUC-ROC of 0.9857 indicates excellent discriminative ability across all classes, with Monocyte achieving a perfect AUC of 1.000 and Lymphocyte achieving 0.999.

The Eosinophil-Neutrophil confusion is the primary limitation. Both cell types exhibit multi-lobed nuclei and similar size ranges, making them challenging to distinguish under certain staining conditions. This is a known difficulty in automated hematology and has been reported in similar studies.

The frozen convolutional backbone strategy proved effective: ImageNet-pretrained low-level features (edges, textures, colour gradients) transfer well to microscopy images. Label smoothing and cosine annealing contributed to stable convergence without overfitting, as evidenced by the close alignment of training and validation curves throughout all 15 epochs.

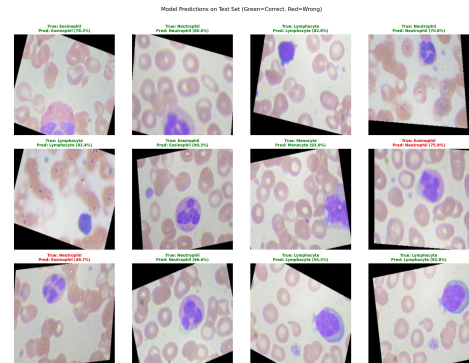


Fig. 5. Sample test predictions. Green = correct, Red = incorrect. Confidence shown.

VI. CONCLUSION AND FUTURE WORK

This paper implemented VGGNet-16 with transfer learning in PyTorch for blood cell classification on the BCCD dataset. A reproducible pipeline with augmentation, cosine LR scheduling, label smoothing, and early stopping was developed. The model achieved 91.63% test accuracy, 0.9160 macro F1-score, and 0.9857 macro AUC-ROC across four cell types.

Future work can improve the system in several directions:

- Unfreeze deeper convolutional layers for end-to-end fine-tuning.
- Apply attention mechanisms or vision transformers.
- Use ensemble methods combining VGGNet-16 with ResNet or EfficientNet.
- Extend to larger or more fine-grained blood cell datasets.

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REFERENCES

- [1] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.
- [3] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [4] G. Litjens et al., "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [5] D. Shen, G. Wu, and H.-I. Suk, "Deep learning in medical image analysis," *Annual Review of Biomedical Engineering*, vol. 19, pp. 221–248, 2017.
- [6] C. Matek, S. Schwarz, K. Spiekermann, and C. Marr, "Human-level recognition of blast cells in acute myeloid leukaemia with convolutional neural networks," *Nature Machine Intelligence*, vol. 1, no. 11, pp. 538–544, 2019.
- [7] P. T. Mooney, "Blood Cell Images," Kaggle, 2018. [Online]. Available: <https://www.kaggle.com/paultimothymooney/blood-cells>
- [8] A. Acevedo et al., "A dataset of microscopic peripheral blood cell images for development of automatic recognition systems," *Data in Brief*, vol. 30, p. 105474, 2020.
- [9] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *Journal of Big Data*, vol. 6, no. 1, p. 60, 2019.
- [10] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.